

Information technology (IT) is the use of computers, storage, networking and other physical devices, [infrastructure](#) and processes to create, process, store, secure and exchange all forms of electronic data. Typically, IT is used in the context of business operations, as opposed to the technology used for personal or entertainment purposes. The commercial use of IT encompasses both computer technology and [telecommunications](#).

Harvard Business Review coined the term *information technology* in 1958 to distinguish between purpose-built machines designed to perform a limited scope of functions and general-purpose computing machines that could be programmed for various tasks. [As the IT industry evolved](#) from the mid-20th century, computing capability increased, while device cost and energy consumption decreased, a cycle that continues today when new technologies emerge.

Types of information technology

Information technology encompasses a wide range of technologies and systems that are used to store, retrieve, process and transmit data for specific use cases.

Common information technology types include the following:

- **Internet and web technologies.** This includes the tools and protocols used to access, navigate and interact with information on the internet. Examples include web browsers, websites, web servers, [Hypertext Markup Language](#), [cascading style sheets](#), [JavaScript](#), [HTTP](#) and other internet-related technologies.
- **Cloud computing.** This involves the delivery of computing resources and services over the internet on a pay-per-use basis. This can include [infrastructure as a service](#), platform as a service, [software as a service](#) and cloud storage options.
- **Databases.** This includes IT systems and software used to store, organize and retrieve data. Examples include [MySQL](#), [NoSQL](#), relational database management systems and [MongoDB](#).
- **Artificial intelligence and machine learning.** [AI](#) and ML-based IT technologies use algorithms and statistical models to enable computers to perform tasks that typically require human intelligence. Examples include speech recognition, image recognition and [natural language processing](#).
- **Cybersecurity.** This type of IT includes technologies and [best practices designed to protect IT systems](#), networks and data from unauthorized access, cyber attacks and other security threats. Cybersecurity can be enforced through firewalls, antivirus software, [encryption](#), intrusion detection systems and security policies.

- **Internet of things.** This includes the network of interconnected devices and sensors that collect, exchange and analyze data. IoT technologies enable the integration of physical objects into computer systems, providing automation, monitoring and control in various domains.
- **IT governance.** This involves making policies and rules for the organization to ensure effective operation.
- **Data analytics and business intelligence.** [BI](#) focuses on tools and techniques for extracting insights from large data sets to support decision-making and business operations. This can include [data mining](#), statistical analysis, data visualization and [predictive modeling](#).

What does information technology encompass?

The IT department ensures that the organization's systems, networks, applications, data and information all connect and function properly. The IT team handles the following three major areas:

THIS ARTICLE IS PART OF

What is IT/OT convergence? Everything you need to know

- Which also includes:
 - [Benefits and challenges of IT/OT convergence](#)
 - [Understand the best IT/OT convergence strategies](#)
 - [5 ways to facilitate the convergence of IT and OT in IoT](#)
1. Deploying and maintaining business applications, services and infrastructure -- including servers, networks and storage.
 2. Monitoring, optimizing and troubleshooting the [performance of applications](#), services and infrastructure.
 3. Overseeing the security and governance of applications, services and infrastructure.

Most IT staff have different responsibilities within the team that can be broken into the following key areas:

- **Administration.** Administrators handle the day-to-day deployment, operation and monitoring of an IT environment, including systems, networks and applications. Admins often perform a range of other duties such as software upgrades, user training, [software license](#) management, procurement, security, data management and observing adherence to business process and compliance requirements. Effective delegation is also part of IT administration and is crucial for a team's productivity.
- **Support.** Help desk staff specialize in answering questions, gathering information and directing troubleshooting efforts for hardware and software. IT support often includes IT asset and change management, helping admins with

procurement, handling backup and recovery of data and applications, monitoring and analyzing logs and other performance monitoring tools and following established support workflows and processes.

- **Applications.** Businesses rely on software to perform work. Some applications, such as email server applications, are procured and deployed by third parties. But many organizations retain a staff of skilled developers that create the applications and interfaces -- such as [application programming interfaces](#) -- needed to deliver critical business capabilities and services. Applications might be coded in a wide array of popular programming languages and integrated with other applications to create smooth and seamless interactions between different applications. Developers might also be tasked with creating interactive business websites and building mobile applications. The trend toward agile or continuous development paradigms requires developers to be increasingly involved with IT operations, such as deploying and monitoring applications.
- **Compliance.** Businesses are obligated to observe varied government and industry-driven regulatory requirements. IT staff play a major role in securing and monitoring access to business data and applications to ensure that such resources are used according to established business governance policy that meets regulatory requirements. Such staff are deeply involved with security tasks and routinely interact with legal and business teams to prevent, detect, investigate and report possible breaches.

IT encompasses many different technologies, capabilities and functions.

Why is information technology important?

It's been said that data is what powers industries worldwide. That might be hyperbole, but few businesses -- large or small -- can remain competitive without the [ability to collect data](#) and turn it into useful information. IT provides the means to develop, process, analyze, exchange, store and secure information.

Data processing plays a significant role in the following core business practices:

- Product development and design.
- Marketing and market research.
- Sales and invoicing.
- Data analysis and decision-making.
- Customer development and retention.
- Accounting and taxes.
- Human resources and payroll.